

# Jinchen Zhao

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## EDUCATION

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| <b>Yale University</b><br>Ph.D. in Applied Physics                                                                        | Aug 2023 – Present<br><i>New Haven, CT</i>        |
| <b>Duke Kunshan University &amp; Duke University</b> (dual degree)<br>B.S. in Applied Mathematics & Computational Science | 2019 – 2023<br><i>Kunshan, China / Durham, NC</i> |

## RESEARCH INTERESTS

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**Quantum Information, Deep Learning, Quantum Optics.** Quantum-Enhanced Physical Neural Networks, Nonlinear Photonics, Quantum Phase Transitions.

## RESEARCH EXPERIENCE

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| <b>Yale University</b><br>Graduate Research Assistant (Advisor: Dr. Logan Wright) | Aug 2023 – Present<br><i>New Haven, CT</i> |
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- Studying unconventional computing platforms (optical, quantum, and thermodynamic), focusing on the role of analog noise in computation at ultra-low energy scales and pathways toward practical, scalable applications.

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| <b>Duke Kunshan University</b><br>Undergraduate Research Assistant (Advisor: Dr. Myung-Joong Hwang) | 2021 – 2023<br><i>Kunshan, China</i> |
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- Discovered the frustration of photons during the superradiant phase transition of a Dicke lattice model and studied its novel critical behaviors.

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| <b>Duke University</b><br>Undergraduate Research Assistant (Advisor: Dr. Kenneth Brown) | Spring 2022<br><i>Durham, NC</i> |
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- Developed an auto-loading module for the trapped ion system, which tested successfully on the device and was added to the official control system.

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| <b>Duke Kunshan University</b><br>Undergraduate Research Assistant (Advisor: Dr. habil. Kai Huang) | 2019 – 2021<br><i>Kunshan, China</i> |
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- Monitored projectile motion under opaque granular material by embedded inertial measurement unit, and studied granular impact under micro-gravity.

## PUBLICATIONS

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- [1] **Jinchen Zhao** and Logan Wright, *Quantum-Enhanced Analog Computing*, in preparation.
- [2] Ali Ansari, Haoran Sun, *et al.* (51 authors, including Logan Wright and **Jinchen Zhao**), *How Good Are Frontier Models at Physics? Expert Re-Grading Reveals Broken Evaluations and Near-Saturation of Leading Benchmarks*, [arXiv:2609.13009](https://arxiv.org/abs/2609.13009) (2026).
- [3] Tivadar Pongó\*, Tianhui Liao\*, **Jinchen Zhao**, Valentin Dichtl, Simeon Völkel, Raúl Cruz Hidalgo, and Kai Huang, *Nature of granular drag in microgravity*, [arXiv:2603.03006](https://arxiv.org/abs/2603.03006) (2026).

- [4] **Jinchen Zhao** and Myung-Joong Hwang, *Anomalous criticality with bounded fluctuations and long-range frustration induced by broken time-reversal symmetry*, *Phys. Rev. Research* 5, L042016 (2023).
- [5] **Jinchen Zhao** and Myung-Joong Hwang, *Frustrated Superradiant Phase Transition*, *Phys. Rev. Lett.* 128, 163601 (2022).
- [6] Stefan Köstler, **Jinchen Zhao**, Chen Lyu, Simeon Völkel, and Kai Huang, *Embedded Inertial Sensor for Tracking Projectile Impact on Granular Media*, *EPJ Web Conf.* 249, 15007 (2021).

## PRESENTATIONS

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- [1] *Tensor Processing with Nonlinear Optical Media*, Frontiers in Optics, 2026.
- [2] *Anomalous Multicritical Phenomena and Frustration Induced by Synthetic Magnetic Fields*, APS March Meeting, 2023.

## INDUSTRY EXPERIENCE

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| <b>NTT Research, Physics &amp; Informatics Lab</b>                                                                                                                                                                                          | Summer 2026   |
| Research Intern                                                                                                                                                                                                                             | <i>Remote</i> |
| <ul style="list-style-type: none"> <li>· Studied the role of quantum optics in optical computing, and tensor computing in programmable nonlinear multimode waveguides.</li> </ul>                                                           |               |
| <b>World Economic Forum</b>                                                                                                                                                                                                                 | 2022          |
| University-Corporation Lab                                                                                                                                                                                                                  |               |
| <ul style="list-style-type: none"> <li>· Led a team of 5 students to map the East Asian quantum computing ecosystem, synthesizing policy, industry, and academic data for the WEF Technology and Quantum Computing Topic Leader.</li> </ul> |               |

## HONORS

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| SPIE Optics and Photonics Scholarship (international scholarship for optics research)    | 2026 |
| Roberts Innovation Fund, Yale Engineering (seed award for research commercialization)    | 2025 |
| Sterling Prize Fellowship, Yale University (top incoming graduate students)              | 2023 |
| Zu Chongzhi Math Student First Award, Duke Kunshan University (top mathematics graduate) | 2023 |
| <i>Magna Cum Laude</i> , Duke Kunshan University                                         | 2023 |
| Graduation with Distinction, Duke Kunshan University                                     | 2023 |

## SERVICE

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- Advisor of 5 undergraduate research projects, Yale University
- Reviewer, *Nature Photonics*
- Reviewer, *Nature Communications*
- Reviewer, *Light: Science & Applications*
- Reviewer, *Journal of the Optical Society of America B*
- Teaching Fellow, APHY 5750 *Physics of AI*, Yale University
- Teaching Fellow, PHYS 180 *University Physics*, Yale University

## SKILLS

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- Programming:** Python (PyTorch, QuTiP, Qiskit), Verilog/HDL
- Hardware:** RFSoc/FPGA (Vivado), RF & photonic instrumentation control
- Tools:** Mathematica, Git, L<sup>A</sup>T<sub>E</sub>X